

**University Of Engineering and Technology, Lahore**  
**Computer Engineering Department**

|                                                                  |                                              |
|------------------------------------------------------------------|----------------------------------------------|
| <b>Course Name: Programming Fundamentals and Data Structures</b> | <b>Course Code: CMPE-112L</b>                |
| <b>Assignment Type: Lab Report</b>                               | <b>Dated:</b>                                |
| <b>Semester: 1st</b>                                             | <b>Session: 2026</b>                         |
| <b>Lab/Project/Assignment #: 8</b>                               | <b>CLOs to be covered: CLO3</b>              |
| <b>Lab Title: Sets, Dictionary</b>                               | <b>Teacher Name: Hafiz Muhammad Mubashar</b> |

**Lab Evaluation:**

| <b>CLO3</b>  | Apply appropriate programming techniques to create executable programs to solve well defined problems |               |               |               |               |               |
|--------------|-------------------------------------------------------------------------------------------------------|---------------|---------------|---------------|---------------|---------------|
|              | <b>Level1</b>                                                                                         | <b>Level2</b> | <b>Level3</b> | <b>Level4</b> | <b>Level5</b> | <b>Level6</b> |
| (5)          |                                                                                                       |               |               |               |               |               |
| <b>Total</b> |                                                                                                       |               |               |               |               | /5            |

**Rubrics for Current Lab Evaluation**

| <b>Scale</b>      | <b>Mark s</b> | <b>Level</b> | <b>Rubric</b>                                                 |
|-------------------|---------------|--------------|---------------------------------------------------------------|
| Excellent         | 5             | L1           | Submitted all lab tasks, BONUS task, have good understanding. |
| Very Good         | 4             | L2           | Submitted the lab tasks but have good understanding           |
| Good              | 3             | L3           | Submitted the lab tasks but have weak understanding.          |
| Basic             | 2             | L4           | Submitted the lab tasks but have no understanding.            |
| Barely Acceptable | 1             | L5           | Submitted only one lab task.                                  |
| Not Acceptable    | 0             | L6           | Did not attempt                                               |

**LAB No 8: Sets, Dictionary**

**Lab Goals/Objectives:**

To understand and use Python's set and dictionary data structures, including their creation, key operations, and set relationship methods (union, intersection, difference, etc.).

**Equipment Required:**

Python 3, VS Code / PyCharm

**Theory:**

A dictionary stores data as key-value pairs and allows fast lookup by key. A set is an unordered collection of unique elements that supports mathematical set operations such as union, intersection, difference, and subset/superset checks.

**Lab Work:****Task 1: Dictionaries**

```
fruits = {"apple": "sayb", "banana": "kayla", "mango": "aam"}  
print(fruits)  
print(fruits["banana"])  
print(fruits["apple"])  
print(fruits["mango"])
```

**Result:****Task 2: Creating a Set**

```
s = {1,2,3,3}  
print(s)
```

**Result:**

```
{1, 2, 3}
```

**Task 3: Set with Mixed Types**

```
s = {1,2,3,"anaya"}  
print(s)
```

**Result:**

```
{3, 1, 'anaya', 2}
```

**Task 4: Empty Set**

```
E =set({})
print(type(E))
```

**Result:**

```
<class 'set'>
```

#### Task 5: Set from a List

```
a = set([1,2,3,4])
print(type(a))
```

**Result:**

```
<class 'set'>
```

#### Task 6: Iterating a Set and Membership Check

```
s={"CE","CS","DS","SE","CE","IST","IT"}
for x in s:
    print(x)
    if "CE" in s:
        print("yes")
```

**Result:**

```
IT
DS
SE
CE
IST
CS
yes
```

#### Task 7: Add and Discard from Set

```
s = {"CE","CS","DS","SE","IST","IT","CE"}
s.add("BE")
print(s)
s.discard("CE")
print(s)
```

**Result:**

```
{'BE', 'IT', 'DS', 'SE', 'CE', 'IST', 'CS'}
{'BE', 'IT', 'DS', 'SE', 'IST', 'CS'}
{'BE', 'IT', 'DS', 'SE', 'IST', 'CS'}
```

**Task 8: Remove from Set**

```
s = {"CE","CS","DS","SE","IST","IT","CE"}  
print(s)  
s.remove("CE")  
print(s)
```

**Result:**

```
{'IT', 'DS', 'SE', 'CE', 'IST', 'CS'}  
{'IT', 'DS', 'SE', 'IST', 'CS'}  
{'IT', 'DS', 'SE', 'IST', 'CS'}
```

**Task 9: Clear a Set**

```
s = {"CE","CS","DS","SE","IST","IT","CE"}  
s.clear()  
print(s)  
print(type(s))
```

**Result:**

```
set()
```

```
<class 'set'>
```

**Task 10: Pop from a Set**

```
s = {"CE","CS","DS","SE","IST","IT","CE"}  
s.pop()  
print(s)  
s.pop()  
print(s)
```

**Result:**

```
{'DS', 'SE', 'CE', 'IST', 'CS'}  
{'SE', 'CE', 'IST', 'CS'}
```

**Task 11: Union and Update**

```
s1 = {10,12,13,14,17}  
s2 = {"anaya","fatima","ushna","fizza","eshaal"}  
s3 = {1,"cat",(1,2,3)}  
print(s.union(s1,s2,s3))  
s1.update(s2)  
print(s1)  
print(s2)
```

**Result:**

```
{1, 10, 'anaya', 12, 13, 14, 'CS', (1, 2, 3), 17, 'fatima',  
  'cat', 'fizza', 'SE', 'CE', 'ushna', 'IST', 'eshaal'}  
{10, 'anaya', 12, 13, 14, 17, 'fatima', 'fizza', 'ushna',  
  'eshaal'}  
{'fatima', 'fizza', 'anaya', 'ushna', 'eshaal'}
```

### Task 12: Intersection

```
s1 = {"UET","NUST","ITU","LUMS"}  
s2 = {"hello","world"}  
print(s1.intersection(s2))
```

**Result:**

```
set()
```

### Task 13: Intersection Update

```
s1 = {1,2,3,4,5,6,7,8,9,10}  
s2 = {10,11,12,13,14,15,16,17,18,19}  
s3 = {10,20,30,40,50,60,70,80,90,100}  
s1.intersection_update(s2)  
print(s1)
```

**Result:**

```
{10}
```

### Task 14: Symmetric Difference

```
s = s1.symmetric_difference(s2)  
print(s)  
print(s1)  
print(s2)
```

**Result:**

```
{11, 12, 13, 14, 15, 16, 17, 18, 19}
```

### Task 15: Disjoint Check

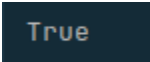
```
s = s1.isdisjoint(s2)  
print(s)
```

**Result:**

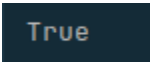
```
False
```

**Task 16: Disjoint Check (Separate Sets)**

```
s1 = {1,2,3,4}
s2 = {5,6,7,8}
s = s1.isdisjoint(s2)
print(s)
```

**Result:**True**Task 17: Subset Check**

```
s1 = {1,2,3,4,5,6,7,8,9,10}
s2 = {2,4,6,8,10}
print(s2.issubset(s1))
```

**Result:**True**Conclusions:**

Dictionaries and sets were explored in detail, including set operations such as union, intersection, difference and subset checking, reinforcing understanding of unordered, unique-element collections.